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Goal-derived information constraints for dual control: Supplementary Material

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S1. Elaboration on proofs in the main paper

S1.1. Bethe Collapse

Proof of Lemma 4 Consider the Bethe free energy functional (4) for the nodes of Fig. 1. The factors correspond to the distributions of the planning model (3) and the beliefs consist of $q_{u,k}$, $q_{x,k}$, $q_{y,k}$, $q_{*,k}$ and q_t and q_θ. Then, it becomes

$$\begin{aligned}\mathcal{B}[q] = &\int q_t \ln \frac{q_t}{p(x_t)} \,\mathrm{d}x_t + \int q_\theta \ln \frac{q_\theta}{p(\theta)} \,\mathrm{d}\theta + \sum_{e \in \mathcal{V}_=} \int q_e \ln \frac{q_e}{f_e} \,\mathrm{d}s_e + \sum_{i \in \mathcal{E}} \mathbb{H}[q_i] \\ &+ \sum_{k \in \tau} \Bigg(\int q_{u,k} \ln \frac{q_{u,k}}{p(u_k)} \,\mathrm{d}u_k + \int q_{x,k} \ln \frac{q_{x,k}}{p(x_k \,|\, x_{k-1}, \theta, u_k)} \,\mathrm{d}s_{x,k} \\ &\qquad + \int q_{y,k} \ln \frac{q_{y,k}}{p(y_k \,|\, x_k)} \,\mathrm{d}s_{y,k} + \int q_{*,k} \ln \frac{q_{*,k}}{p(y_k \,|\, \phi_*)} \,\mathrm{d}y_k \Bigg). \end{aligned} \tag{S1}$$

The marginalization constraints (**??**) give every node with a single edge the belief of that edge, so $q_t = q(x_t)$, $q_\theta = q(\theta)$, $q_{u,k} = q(u_k)$ and $q_{*,k} = q(y_k)$. Every equality node confines its belief to equal values on its edges, $q_e = q(s)\,\delta(s - s')\,\delta(s - s'')$, and the marginalization constraints make $q(s)$ the belief of all three edges. All copies of θ and x_k therefore carry the beliefs $q(\theta)$ and $q(x_k)$, and the deltas cancel in the log-ratio of every equality node. The form constraints (Definition 2) pin $q(x_t) = p(x_t)$ and $q(\theta) = p(\theta)$, which removes the first two terms. The goal terms split into an entropy and a cross-entropy, so

$$\begin{aligned}\mathcal{B}[q] = &\sum_{k \in \tau} \Bigg(D_{\mathrm{KL}}\big[q(u_k) \,\|\, p(u_k)\big] + \int q_{x,k} \ln \frac{q_{x,k}}{p(x_k \,|\, x_{k-1}, \theta, u_k)} \,\mathrm{d}s_{x,k} \\ &\qquad + \int q_{y,k} \ln \frac{q_{y,k}}{p(y_k \,|\, x_k)} \,\mathrm{d}s_{y,k} - \mathbb{H}\big[q(y_k)\big] - \mathbb{E}_{q(y_k)}\big[\ln p(y_k \,|\, \phi_*)\big] \Bigg) \\ &- \sum_{e \in \mathcal{V}_=} \mathbb{H}\big[q_{\mathrm{in}(e)}\big] + \sum_{i \in \mathcal{E}} \mathbb{H}[q_i]\,, \end{aligned} \tag{S2}$$

where $q_{\mathrm{in}(e)}$ is the belief of the input edge of e. The factorization constraints (Definition 1) set $q_{x,k} = p(x_k \,|\, x_{k-1}, \theta, u_k)\, q(x_{k-1})\, q(\theta)\, q(u_k)$ and $q_{y,k} = p(y_k \,|\, x_k)\, q(x_k)$, so the conditional factors cancel in the log-ratios. These factors are normalised densities of the outputs x_k and y_k, so integrating

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out the outputs leaves the input beliefs. The logarithm of their product then splits into the negative entropies of the inputs,

$$\begin{aligned}\mathcal{B}[q] &= \sum_{k\in\tau}\Big(D_{\mathrm{KL}}\big[q(u_k)\,\|\,p(u_k)\big] + \int q(x_{k-1})\,q(\theta)\,q(u_k)\ln\big(q(x_{k-1})\,q(\theta)\,q(u_k)\big)\,\mathrm{d}(x_{k-1},\theta,u_k)\\ &\quad + \int q(x_k)\ln q(x_k)\,\mathrm{d}x_k - \mathbb{H}\big[q(y_k)\big] - \mathbb{E}_{q(y_k)}\big[\ln p(y_k\,|\,\phi_*)\big]\Big) - \sum_{e\in\mathcal{V}_=}\mathbb{H}\big[q_{\mathrm{in}(e)}\big] + \sum_{i\in\mathcal{E}}\mathbb{H}[q_i]\\ &= \sum_{k\in\tau}\Big(D_{\mathrm{KL}}\big[q(u_k)\,\|\,p(u_k)\big] - \mathbb{H}\big[q(x_{k-1})\big] - \mathbb{H}\big[q(\theta)\big] - \mathbb{H}\big[q(u_k)\big] - \mathbb{H}\big[q(x_k)\big]\\ &\qquad - \mathbb{H}\big[q(y_k)\big] - \mathbb{E}_{q(y_k)}\big[\ln p(y_k\,|\,\phi_*)\big]\Big) - \sum_{e\in\mathcal{V}_=}\mathbb{H}\big[q_{\mathrm{in}(e)}\big] + \sum_{i\in\mathcal{E}}\mathbb{H}[q_i]\,. \end{aligned} \tag{S3}$$

We orient every edge along the generative model (3), from the node that generates its variable to the node that consumes it. The belief nodes and the control priors then have no inputs, and every edge is the input of exactly one transition, measurement, equality or goal node. These nodes have the inputs (x_{k-1},θ,u_k), x_k, $\mathrm{in}(e)$ and y_k, so the edge entropies sum to

$$\sum_{i\in\mathcal{E}}\mathbb{H}[q_i] = \sum_{k\in\tau}\Big(\mathbb{H}\big[q(x_{k-1})\big] + \mathbb{H}\big[q(\theta)\big] + \mathbb{H}\big[q(u_k)\big] + \mathbb{H}\big[q(x_k)\big] + \mathbb{H}\big[q(y_k)\big]\Big) + \sum_{e\in\mathcal{V}_=}\mathbb{H}\big[q_{\mathrm{in}(e)}\big]\,. \tag{S4}$$

Substituting (S4) into (S3) cancels every entropy and leaves

$$\mathcal{B}[q] = \sum_{k\in\tau}\Big(D_{\mathrm{KL}}\big[q(u_k)\,\|\,p(u_k)\big] - \mathbb{E}_{q(y_k)}\big[\ln p(y_k\,|\,\phi_*)\big]\Big)\,, \tag{S5}$$

which is (12). The integration over the outputs also shows that the marginals of $q_{x,k}$ and $q_{y,k}$ on their input edges equal the edge beliefs. The node beliefs thus meet the marginalization constraints (**??**) at their inputs without multipliers, as Section 4 uses. ■

S1.2. Control Posterior

Theorem 5 follows from the stationarity conditions of the Lagrangian (13). The multipliers of the output and state constraints obey a backward recursion that starts from the goal log-densities, and this recursion is the backward pass of message passing. Every input belief then takes a Gibbs form, whose exponent sums the expected goal log-densities of the later outputs and, for the next input, the weighted information gain.

Proof of Theorem 5 *Free beliefs.* The form constraints fix $q(x_t)$ and $q(\theta)$, and Lemma 4 expresses the Bethe free energy through the edge beliefs. The Lagrangian (13) is therefore a functional of the beliefs $q(u_k)$, $q(x_k)$ and $q(y_k)$ for $k\in\tau$. The joint belief (8) holds only factors and the fixed beliefs $q(\theta)$ and $q(x_t)$, so the information gain $\mathbb{I}_{t+1}(u)$ at an input value u depends on no free belief. The information gain of the input belief is therefore linear in $q(u_{t+1})$,

$$\mathbb{I}_{t+1}[q] = \int_{\mathcal{U}} q(u_{t+1})\,\mathbb{I}_{t+1}(u_{t+1})\,\mathrm{d}u_{t+1}\,. \tag{S6}$$

At a stationary point, the functional derivative of $\mathcal{L}$ with respect to every free belief vanishes. We evaluate these derivatives in the order of the backward pass, from the outputs to the states to the inputs.

Outputs. The belief $q(y_k)$ enters only the goal term $-\mathbb{E}_{q(y_k)}[\ln p(y_k \mid \phi_*)]$ of (12) and the marginalization constraint of y_k in (13). Its functional derivative vanishes if

$$\frac{\delta \mathcal{L}}{\delta q(y_k)} = -\ln p(y_k \mid \phi_*) + \lambda_{y_k}(y_k) = 0\,, \qquad \text{that is,} \qquad \lambda_{y_k}(y_k) = \ln p(y_k \mid \phi_*)\,. \tag{S7}$$

The multiplier of every output is thus the log-message $\ln p(y \mid \phi_*)$ of its goal node.

States. The belief $q(x_k)$ enters three terms of (13). These are its own marginalization constraint, the constraint of y_k through the measurement factor and, for $k < t+K$, the constraint of x_{k+1} through the next transition factor. Its functional derivative is

$$\begin{aligned}\frac{\delta \mathcal{L}}{\delta q(x_k)} = \lambda_{x_k}(x_k) &- \int \lambda_{y_k}(y_k)\, p(y_k \mid x_k)\, \mathrm{d}y_k \\ &- \int \lambda_{x_{k+1}}(x_{k+1})\, p(x_{k+1} \mid x_k, \theta, u_{k+1})\, q(\theta)\, q(u_{k+1})\, \mathrm{d}(x_{k+1}, \theta, u_{k+1})\,,\end{aligned} \tag{S8}$$

without the last term for $k = t+K$. Setting (S8) to zero gives the recursion (22). With $\nu_{d_l \to l} = \exp(\lambda_l)$, its two terms are the log-messages (20) that the measurement node and the next transition node send to x_k, and the equality node of x_k multiplies these messages. The recursion is therefore the backward pass of Section 4.3.

Unrolling. The recursion expresses the multiplier of every state through the goal log-densities of the later outputs. For an input value u_k, the marginalization constraints propagate the predictions

$$\begin{aligned} q(x_k \mid u_k) &= \int p(x_k \mid x_{k-1}, \theta, u_k)\, q(x_{k-1})\, q(\theta)\, \mathrm{d}(x_{k-1}, \theta)\,, \\ q(x_m \mid u_k) &= \int p(x_m \mid x_{m-1}, \theta, u_m)\, q(x_{m-1} \mid u_k)\, q(\theta)\, q(u_m)\, \mathrm{d}(x_{m-1}, \theta, u_m)\,, \\ q(y_m \mid u_k) &= \int p(y_m \mid x_m)\, q(x_m \mid u_k)\, \mathrm{d}x_m\,, \end{aligned} \tag{S9}$$

with $m = k+1, \ldots, t+K$ in the second line and $m = k, \ldots, t+K$ in the third. The next transition belief maps $q(x_m \mid u_k)$ to $q(x_{m+1} \mid u_k)$, and the measurement factor maps it to $q(y_m \mid u_k)$. Taking the expectation of (22) under $q(x_m \mid u_k)$ and substituting (S7) therefore gives

$$\mathbb{E}_{q(x_m \mid u_k)}\big[\lambda_{x_m}(x_m)\big] = \mathbb{E}_{q(y_m \mid u_k)}\big[\ln p(y_m \mid \phi_*)\big] + \mathbb{E}_{q(x_{m+1} \mid u_k)}\big[\lambda_{x_{m+1}}(x_{m+1})\big]\,, \tag{S10}$$

without the last term for $m = t+K$. Applying (S10) from $m = k$ to $m = t+K$ leaves

$$\mathbb{E}_{q(x_k \mid u_k)}\big[\lambda_{x_k}(x_k)\big] = \sum_{m=k}^{t+K} \mathbb{E}_{q(y_m \mid u_k)}\big[\ln p(y_m \mid \phi_*)\big]\,. \tag{S11}$$

Inputs. The belief $q(u_k)$ enters four terms of (13). These are its divergence from the prior, its normalization, the constraint of x_k through the transition node b that consumes u_k and, for $k = t+1$, the information term (S6). Its functional derivative is

$$\frac{\delta \mathcal{L}}{\delta q(u_k)} = \ln \frac{q(u_k)}{p(u_k)} + 1 + \psi_k - \mathbb{E}_{q_b(s_b \mid u_k)}\big[\lambda_{x_k}(x_k)\big] - \gamma_k\, \mathbb{I}_k(u_k)\,, \tag{S12}$$

with $\gamma_{t+1} = \gamma$, $\gamma_k = 0$ for $k > t+1$, and the belief $q_b(s_b \mid u_k) = p(x_k \mid x_{k-1}, \theta, u_k)\, q(x_{k-1})\, q(\theta)$ of b at a fixed u_k. Integrating q_b over x_{k-1} and θ leaves $q(x_k \mid u_k)$, so the expectation in (S12) is the left-hand side of (S11). Setting (S12) to zero and substituting (S11) gives

$$q^\star(u_k) = p(u_k) \exp\Big(\sum_{m=k}^{t+K} \mathbb{E}_{q(y_m \mid u_k)} \big[\ln p(y_m \mid \phi_*) \big] + \gamma_k\, \mathbb{I}_k(u_k) \Big)\, e^{-1-\psi_k} \,. \tag{S13}$$

The multiplier ψ_k enters only through the constant $e^{-1-\psi_k}$, which the normalization of $q(u_k)$ fixes. With $p(y_m \mid \phi_*) = p(y_m \mid \phi_*)$, (S13) is the posterior (14), and its Gibbs form is that of control as inference (Levine, 2018). ■

S1.3. Information Multiplier

Lemma 6 follows from the log-normaliser of the posterior of the next input, which is an exponential family in the multiplier. The first derivative of the log-normaliser is the expected information gain $h(\gamma)$, and its second derivative is the variance of the information gain, which is positive. As the multiplier grows, the posterior concentrates on the most informative inputs, so that $h(\gamma)$ rises towards $\mathbb{I}^{\max}$. The KKT conditions (16) therefore admit exactly one multiplier whenever the floor lies below $\mathbb{I}^{\max}$.

Proof of Lemma 6 *Exponential family.* We fix all beliefs except that of the next input, and we write u for the value of u_{t+1} and $\mathbb{I}(u)$ for its information gain $\mathbb{I}_{t+1}(u)$. For $k = t+1$, Theorem 5 gives the posterior

$$q^\star(u; \gamma) = \frac{p(u)\, \nu_0(u)\, e^{\gamma \mathbb{I}(u)}}{Z(\gamma)} \,, \qquad Z(\gamma) = \int_{\mathcal{U}} p(u)\, \nu_0(u)\, e^{\gamma \mathbb{I}(u)}\, \mathrm{d}u \,, \tag{S14}$$

where the message

$$\nu_0(u) = \exp\Big(\sum_{m=t+1}^{t+K} \mathbb{E}_{q(y_m \mid u)} \big[\ln p(y_m \mid \phi_*) \big] \Big) \tag{S15}$$

collects the expected goal log-densities of the K outputs of the horizon. The predictions $q(y_m \mid u)$ depend only on the input value u and on the fixed beliefs. The information gain depends only on u as well, because the joint belief (8) holds only factors and the fixed beliefs $q(\theta)$ and $q(x_t)$. Hence neither ν_0 nor $\mathbb{I}$ depends on γ, and the posterior (S14) is an exponential family (Wainwright et al., 2008). Its natural parameter is γ, its sufficient statistic is $\mathbb{I}(u)$, its base measure is $p(u)\, \nu_0(u)$, and its log-normaliser is $A(\gamma) = \ln Z(\gamma)$. By (S6), the constraint value is linear in the input belief, so at the posterior it takes the value

$$\mathbb{I}_{t+1}[q^\star] = \int_{\mathcal{U}} \mathbb{I}(u)\, q^\star(u; \gamma)\, \mathrm{d}u = h(\gamma) \,. \tag{S16}$$

Normaliser. The normaliser $Z(\gamma)$ is finite and positive for every $\gamma \geq 0$, because the prior is normalised and the other two factors of the integrand are bounded. The information gain is a mutual information, so it is nonnegative (Cover and Thomas, 1999), and by assumption it has an upper bound M,

$$0 \leq \mathbb{I}(u) \leq M \qquad \text{for all } u \in \mathcal{U} \,. \tag{S17}$$

The Gaussian goal prior has the largest density $c_* = (2\pi)^{-n_y/2}\,|S_*|^{-1/2}$, so every expected goal log-density in (S15) is at most $\ln c_*$. Every such expectation is also finite by (15), because the predictions have finite means and covariances. The message thus satisfies $0 < \nu_0(u) \leq c_*^K$, and since the prior integrates to one,

$$0 < Z(\gamma) \leq c_*^K\, e^{\gamma M} \int_{\mathcal{U}} p(u)\,\mathrm{d}u = c_*^K\, e^{\gamma M}\,. \tag{S18}$$

The lower bound holds because the integrand is positive wherever $p(u) > 0$.

Derivatives. The first two derivatives of the log-normaliser are the mean and the variance of the information gain under the posterior. For $|\gamma' - \gamma| \leq 1$ and $j \in \{1, 2\}$, the integrable function $c_*^K M^j e^{(\gamma+1)M} p(u)$ dominates the integrands $p(u)\,\nu_0(u)\,\mathbb{I}(u)^j e^{\gamma'\mathbb{I}(u)}$, so we differentiate Z under the integral sign. The first derivative of A is

$$A'(\gamma) = \frac{Z'(\gamma)}{Z(\gamma)} = \int_{\mathcal{U}} \mathbb{I}(u)\,\frac{p(u)\,\nu_0(u)\,e^{\gamma\mathbb{I}(u)}}{Z(\gamma)}\,\mathrm{d}u = \mathbb{E}_{q^\star(u;\gamma)}\big[\mathbb{I}(u)\big] = h(\gamma)\,, \tag{S19}$$

and the second derivative is

$$\begin{aligned} A''(\gamma) &= \frac{Z''(\gamma)}{Z(\gamma)} - \left(\frac{Z'(\gamma)}{Z(\gamma)}\right)^2 = \mathbb{E}_{q^\star(u;\gamma)}\big[\mathbb{I}(u)^2\big] - \mathbb{E}_{q^\star(u;\gamma)}\big[\mathbb{I}(u)\big]^2 \\ &= \mathbb{V}_{q^\star(u;\gamma)}\big[\mathbb{I}(u)\big]\,. \end{aligned} \tag{S20}$$

The function $h = A'$ is thus differentiable, and therefore continuous, with the derivative $h'(\gamma) = \mathbb{V}_{q^\star(u;\gamma)}[\mathbb{I}(u)]$.

Monotonicity. The variance (S20) is positive, so h is strictly increasing. A variance vanishes only if its variable is almost surely constant. The posterior (S14) is the prior times the positive weight $\nu_0(u)\,e^{\gamma\mathbb{I}(u)}/Z(\gamma)$, so a set of inputs has posterior probability zero if and only if it has prior probability zero. The information gain is not $p(u)$-almost surely constant, so it is not $q^\star(u;\gamma)$-almost surely constant either, and $h'(\gamma) > 0$ for every $\gamma \geq 0$.

Laplace principle. As the multiplier grows, the posterior moves its mass to the most informative inputs, so that $h(\gamma) \to \mathbb{I}^{\max}$. By its definition (18), the essential supremum $\mathbb{I}^{\max}$ is the smallest c with $\mathbb{I}(u) \leq c$ for $p(u)$-almost every u. It is positive, because $\mathbb{I}$ is nonnegative and not almost surely constant. The posterior shares its null sets with the prior, so $\mathbb{I}(u) \leq \mathbb{I}^{\max}$ holds $q^\star$-almost surely, and $h(\gamma) \leq \mathbb{I}^{\max}$.

For a lower bound, we fix $0 < \epsilon < \mathbb{I}^{\max}/2$ and let $\mathcal{U}_\delta = \{u \in \mathcal{U} : \mathbb{I}(u) > \mathbb{I}^{\max} - \delta\}$ be the set of inputs whose information gain lies within δ of the supremum. The prior gives $\mathcal{U}_\epsilon$ a positive probability, because $\mathbb{I}^{\max} - \epsilon$ lies below the essential supremum. Since $\nu_0 > 0$, the set $\mathcal{U}_\epsilon$ also has the positive weight $m_\epsilon = \int_{\mathcal{U}_\epsilon} p(u)\,\nu_0(u)\,\mathrm{d}u$, and restricting Z to $\mathcal{U}_\epsilon$ gives

$$Z(\gamma) \geq \int_{\mathcal{U}_\epsilon} p(u)\,\nu_0(u)\,e^{\gamma\mathbb{I}(u)}\,\mathrm{d}u \geq m_\epsilon\, e^{\gamma(\mathbb{I}^{\max}-\epsilon)}\,. \tag{S21}$$

Outside $\mathcal{U}_{2\epsilon}$, the information gain is at most $\mathbb{I}^{\max} - 2\epsilon$. With $\nu_0 \leq c_*^K$ and (S21), the posterior probability of these inputs is therefore at most

$$\int_{\mathcal{U}\setminus\mathcal{U}_{2\epsilon}} q^\star(u;\gamma)\,\mathrm{d}u \leq \frac{c_*^K\, e^{\gamma(\mathbb{I}^{\max}-2\epsilon)}}{Z(\gamma)} \leq \frac{c_*^K}{m_\epsilon}\, e^{-\gamma\epsilon}\,, \tag{S22}$$

which vanishes as $\gamma \to \infty$. On $\mathcal{U}_{2\epsilon}$, the information gain exceeds $\mathbb{I}^{\max} - 2\epsilon > 0$, and on the rest of $\mathcal{U}$, it is nonnegative by (S17). We split the expectation $h(\gamma)$ over these two sets and obtain

$$h(\gamma) \geq \big(\mathbb{I}^{\max} - 2\epsilon\big) \int_{\mathcal{U}_{2\epsilon}} q^\star(u;\gamma)\,\mathrm{d}u \geq \big(\mathbb{I}^{\max} - 2\epsilon\big)\Big(1 - \frac{c_*^K}{m_\epsilon}\, e^{-\gamma\epsilon}\Big)\,. \tag{S23}$$

The function h is increasing and bounded by $\mathbb{I}^{\max}$, so it has a limit, and (S23) places this limit at or above $\mathbb{I}^{\max} - 2\epsilon$ for every ϵ. Hence $h(\gamma) \to \mathbb{I}^{\max}$ as $\gamma \to \infty$. Since h is strictly increasing, it also stays below its limit, with $h(\gamma) < h(\gamma + 1) \leq \mathbb{I}^{\max}$ for every $\gamma \geq 0$.

Cases. By (S16), the KKT conditions (16) constrain the multiplier alone,

$$\gamma \geq 0\,, \qquad h(\gamma) \geq \beta_t\,, \qquad \gamma\big(\beta_t - h(\gamma)\big) = 0\,. \tag{S24}$$

Complementary slackness admits a positive multiplier only at a root of $h(\gamma) = \beta_t$. Which multiplier meets all three conditions depends on where the floor lies relative to $h(0)$ and $\mathbb{I}^{\max}$.

If $\beta_t \leq h(0)$, then $\gamma = 0$ meets all three conditions. Every $\gamma > 0$ gives $h(\gamma) > h(0) \geq \beta_t$, so the product $\gamma(\beta_t - h(\gamma))$ is negative and violates complementary slackness. The multiplier $\gamma = 0$ is therefore the only one.

If $h(0) < \beta_t < \mathbb{I}^{\max}$, then $\gamma = 0$ violates primal feasibility, so the multiplier is positive and solves $h(\gamma) = \beta_t$. Because $h(\gamma) \to \mathbb{I}^{\max} > \beta_t$, some $\bar\gamma > 0$ has $h(\bar\gamma) > \beta_t$. The function h is continuous with $h(0) < \beta_t < h(\bar\gamma)$, so the intermediate value theorem gives a root in $(0, \bar\gamma)$, and strict monotonicity makes this root unique.

If $\beta_t \geq \mathbb{I}^{\max}$, then $h(\gamma) < \mathbb{I}^{\max} \leq \beta_t$ for every $\gamma \geq 0$, so no multiplier meets primal feasibility. For $\beta_t > \mathbb{I}^{\max}$, no belief meets the constraint at all. Every belief with a finite divergence from the prior puts no mass where the prior puts none, so its expected information gain is at most $\mathbb{I}^{\max}$. ■

S2. Details of main experiment

This section specifies the quadcopter experiment of Section 5: the simulator, the distributions of the planning model, the approximations and the messages of the planner, the baselines and the protocol.

S2.1. Simulator

The simulator draws a flight from its seed and integrates the dynamics with the drawn parameters. Every flight starts at rest at $p_0 = (0, 0, 2)$ m, and its target $y_* = p_0 + (5\cos h, 5\sin h, 1)$ m has the heading $h \sim \mathcal{U}[0, 2\pi)$. Within a step of $\Delta = 0.5$ s, an inner loop tracks the commanded thrust per mass $\alpha_k = g\,(1 + u_{k,1}/2)$, the roll $\varphi = 0.4\,u_{k,2}$ and the pitch $\vartheta = 0.4\,u_{k,3}$, with zero yaw. The drag coefficient $k \sim \mathcal{U}[0.025, 0.055]$ s/m and the heading fault $\chi \sim \mathcal{U}[-2.6, 2.6]$ rad stay constant over the flight.

The fault rotates the whole thrust axis about the vertical, $\tilde n_k = R_z(\chi)\,n_k$, so the simulator computes the specific force

$$f_k = \alpha_k \tilde n_k + \alpha_k \tilde\Pi_k\big(k\,\omega_k - k\,\dot p_{k-1}\big) + w_k\,, \qquad \tilde\Pi_k = I - \tilde n_k \tilde n_k^\top\,, \tag{S25}$$

with the wind ω_k and the unmodelled force $w_k \sim \mathcal{N}(0, 0.2^2 I)$ m/s^2. Since $R_z(\chi)\,n_k = (\Gamma n_k^{\mathrm{h}}, n_k^{\mathrm{v}})$ for $(a, b) = (\cos\chi, \sin\chi)$, the thrust term equals that of (21) with $\eta = 1$. The drag, however,

acts in the rotated rotor plane $\tilde{\Pi}_k$, whereas the model keeps the commanded rotor plane Π_k, so the model is misspecified whenever the quadcopter tilts. The position fix has the noise $\mathcal{N}(0, 0.3^2 I)$ m^2, and the IMU noise has the standard deviation $\sigma_f \in \{0.2, 0.5, 1, 2, 3\}$ m/s^2, with the nominal value $\sigma_f = 2$ m/s^2.

The wind varies smoothly, with a smoothness that differs between seeds, whereas the model treats the wind drag $c_k = k\,\omega_k$ as a random walk. The mean wind speed follows a Weibull distribution with shape 2 and scale 5 m/s (Justus et al., 1978), clipped to $[0.5, 8]$ m/s, and the mean direction is uniform. Stationary Ornstein–Uhlenbeck processes perturb the speed with a standard deviation of 20% of the mean speed, the direction with 0.35 rad and the vertical wind with 0.3 m/s. Their three correlation times are drawn log-uniformly from $[1, 10]$ s. The cap of 8 m/s bounds the mean wind, but gusts on a vehicle with a large drag coefficient can still push the quadcopter off target, and Section S2.5 drops the seeds on which even the oracle fails.

S2.2. Planning model

The planning model instantiates (3) with Gaussian beliefs, a Gaussian transition and likelihood, a Beta control prior and a Gaussian goal prior on the position (Table S1). The state $x_k = (p_k, \dot{p}_k, f_k) \in \mathbb{R}^9$ contains the specific force, so that the IMU measures a state, and the parameter vector $\theta = (c, k, a, b, \eta) \in \mathbb{R}^7$ contains the current wind drag c. The transition mean $F(x_{k-1}, \theta, u_k)$ is the right-hand side of (21) without w_k. The unmodelled force enters the position, the velocity and the force through $L = [\frac{1}{2}\Delta^2 I;\ \Delta I;\ I] \in \mathbb{R}^{9\times 3}$, which gives the process noise covariance $Q_x = 0.2^2\, LL^\top$. The control prior is the Beta$(2, 2)$ density of $(u_{k,i} + 1)/2$ for every input component, which vanishes at the actuator limits and keeps the plans inside $\mathcal{U}$.

Table S1: Factors of the planning model (3) for the quadcopter, at the step $k \in \tau$ of the horizon. The filter supplies m_t, S_t, $\hat{\theta}_t$ and P_t at every time t.

Factor	Distribution
State belief $p(x_t)$	$\mathcal{N}(m_t, S_t)$, the state marginal of the filter
Parameter belief $p(\theta)$	$\mathcal{N}\big(\hat{\theta}_t, P_t + (k-t)\,W\big)$, from the parameter marginal of the filter
Transition $p(x_k \mid x_{k-1}, \theta, u_k)$	$\mathcal{N}\big(F(x_{k-1}, \theta, u_k), Q_x\big)$
Likelihood $p(y_k \mid x_k)$	$\mathcal{N}\big((p_k, f_k), \mathrm{diag}(0.3^2 I, \sigma_f^2 I)\big)$
Control prior $p(u_k)$	$\prod_{i=1}^{3} \frac{3}{4}\,(1 - u_{k,i}^2)$ on $\mathcal{U} = [-1, 1]^3$
Goal prior $p(y \mid \phi_*)$	$\mathcal{N}(y_*, S_*)$ on the position output, with $S_* = (0.03\ \mathrm{m})^2 I$

The priors at $t = 0$ state the spread of the seed distribution, because the floor sees only the uncertainty that the beliefs state. The state prior is $\mathcal{N}\big((p_0, 0, ge_3), \mathrm{diag}(0.3^2 I, 0.05^2 I, I)\big)$, and the filter conditions on the first position fix before the first plan. The wind drag has the prior $\mathcal{N}\big(0, \mathrm{diag}(s_h^2, s_h^2, s_v^2)\big)$ with $s_v = 0.016$ and $s_h = 0.04 \cdot 5/\sqrt{2} = 0.14$, which matches $\bar{k}^2\, \mathbb{E}[\omega_x^2]$ for the mean drag coefficient $\bar{k} = 0.04$ s/m under the Weibull wind. The drag coefficient has the prior $\mathcal{N}(0.04, 0.01^2)$, and the vertical gain has the prior $\mathcal{N}(1, 0.05^2)$. The pair (a, b) has the prior $\mathcal{N}\big((0.20, 0), \mathrm{diag}(0.61^2, 0.76^2)\big)$, which matches the first two moments of $(\cos\chi, \sin\chi)$ under the fault distribution.

Between two steps, the wind drag follows a random walk with the steps $\mathcal{N}(0, 0.025^2 I)$, because the wind varies, and the gains (a, b, η) follow a random walk with the steps $\mathcal{N}(0, 0.02^2 I)$, which keeps their estimates adaptive. The drag coefficient k stays constant. Over the horizon, these random walks widen the parameter belief by $W = \mathrm{diag}(0.025^2 I, 0, 0.02^2 I)$ per step (Table S1).

S2.3. Approximations and message passing

Both the filter and the planner linearise the dynamics, and neither uses an unscented transform or sampling. The filter is an extended Kalman filter over the joint vector (x_k, θ), which linearises F at the posterior mean with Jacobians that CasADi computes by automatic differentiation. Given the input, F is affine in θ for a fixed state and affine in the state for fixed parameters, and only the product $k\,\dot{p}_{k-1}$ in the drag couples the two. The linearisation therefore drops only the second-order term of this product. The measurement update is exact given the Gaussian prediction, because the outputs select p_k and f_k linearly.

The planner passes the messages of Fig. 1 over the horizon and keeps every control belief at its mode u_k, which the controller applies. The parameter message ④ of step k is $\mathcal{N}(\hat{\theta}_t, P_k)$ with $P_k = P_t + (k-t)\,W$, and the forward messages ③ and ⑤ propagate the state belief from $\mathcal{N}(m_t, S_t)$ through the transition nodes, linearised at their incoming means,

$$m_k = F(m_{k-1}, \hat{\theta}_t, u_k)\,, \qquad S_k = A_k S_{k-1} A_k^\top + B_k P_k B_k^\top + Q_x\,, \tag{S26}$$

with the Jacobians $A_k = \partial F/\partial x$ and $B_k = \partial F/\partial \theta$ at $(m_{k-1}, \hat{\theta}_t, u_k)$. The forward message ⑦ of the position output then has the mean $\mu_k = m_k^{\mathrm{p}}$ and the covariance $\Sigma_k = S_k^{\mathrm{pp}} + 0.3^2 I$, which enter the cross-entropy (15).

Three approximations shape (S26). First, the covariance drops the product of the uncertainties of k and $\dot{p}_{k-1}$, which the linearisation neglects, whereas the mean is exact for independent x_{k-1} and θ. Second, the factorization constraint of Definition 1 draws x_{k-1} and θ independently in every transition. The prediction (S26) thus adds the parameter uncertainty afresh at every step instead of propagating a persistent parameter error, and it drops the cross-covariance between x_t and θ that the filter holds. Third, the beliefs stay fixed within the plan, as in explicit dual control, so the parameter covariance grows by the random walks and does not shrink through anticipated measurements.

The information and its floor use the position output, because the goal acts only on the position. The next position output carries the information $\mathbb{I}_{t+1}(u) = \frac{1}{2} \ln \left| \Sigma_{t+1}(u)\,(0.3^2 I)^{-1} \right|$ about x_{t+1} and θ, which is (7) for the Gaussian joint belief (8). The floor (11) takes $R = 0.3^2 I$ and replaces the expectation in (10) by a self-normalised average over 4096 inputs that are drawn uniformly from $(-1, 1)^3$ with a fixed seed, so that every step and every controller share these nodes. At every node, one step of (S26) gives $\mu(u)$ and $\Sigma(u)$. The floor is capped at the largest information on the nodes minus 0.01, which approximates $\mathbb{I}^{\max}$ of (18) and keeps the multiplier of Lemma 6 finite.

The backward messages ⑥ carry the free energy to go of the plan, which depends on the forward message $\mathcal{N}(m, S)$ of x_k as $v_k(m) + \frac{1}{2}\mathrm{tr}(\Omega_k S)$, up to a constant. With the matrix H that selects the position and $G = H^\top S_*^{-1} H$, the measurement and goal nodes of x_k contribute $\frac{1}{2}\|Hx - y_*\|^2_{S_*^{-1}}$ and the weight G. The equality node of x_k adds these to the message that the transition node of u_{k+1} maps back from x_{k+1},

$$\begin{aligned} v_k(x) &= \tfrac{1}{2}\|Hx - y_*\|^2_{S_*^{-1}} + v_{k+1}\big(F(x, \hat{\theta}_t, u_{k+1})\big) + \tfrac{1}{2}\mathrm{tr}\big(\Omega_{k+1} B_{k+1}(x) P_{k+1} B_{k+1}(x)^\top\big)\,, \\ \Omega_k &= G + A_{k+1}^\top \Omega_{k+1} A_{k+1}\,, \end{aligned} \tag{S27}$$

from $v_{t+K}(x) = \frac{1}{2}\|Hx - y_*\|^2_{S_*^{-1}}$ and $\Omega_{t+K} = G$. The node is affine in the state for fixed parameters, and its Jacobian $B_{k+1}(x)$ is affine in the state, because the column of k grows with $\dot{p}$. Every v_k is therefore exactly quadratic, and the planner stores it in information form. The message ② of the transition node to its control edge evaluates the free energy to go at the forward message of an input value u,

$$-\ln \nu_k(u) = v_k\big(F(m_{k-1}, \hat{\theta}_t, u)\big) + \tfrac{1}{2}\mathrm{tr}\big(\Omega_k S_k(u)\big)\,, \tag{S28}$$

where $S_k(u)$ is the covariance of (S26) at the input value u.

By Theorem 5, the mode of the control belief maximises $p(u)\,\nu_k(u)\,e^{\gamma_k \mathbb{I}_k(u)}$, with $\gamma_{t+1} = \gamma$ and $\gamma_k = 0$ for later inputs. Damped Newton steps find this mode from the current input, with the gradient and the Hessian of (S28) from CasADi, mirrored negative curvature and a backtracking line search inside $(-1, 1)^3$. A sweep runs a forward pass of (S26) and then the backward pass of (S27), which updates every input on its way from u_{t+K} to u_{t+1}. When the pass updates u_k, the earlier inputs have not changed since the forward pass, and the backward message already carries the updated later inputs. Every update therefore minimises the negative log-posterior of the plan,

$$J(u_\tau) = \sum_{k\in\tau}\Big(-\sum_{i=1}^{3}\ln\big(1-u_{k,i}^2\big) + \tfrac{1}{2}\|\mu_k - y_*\|^2_{S_*^{-1}} + \tfrac{1}{2}\mathrm{tr}\big(S_*^{-1}\Sigma_k\big)\Big) - \gamma\,\mathbb{I}_{t+1}(u_{t+1})\,, \tag{S29}$$

over one input, given the others. This objective is the negative logarithm of the control priors, the exponentiated cross-entropies (15) and the information term of (14), up to a constant. The sweeps thus decrease J monotonically, and they stop once no input moves by more than 10^{-7}. A fixed point minimises J over every input given the others, so it is a stationary point of J. The planner starts from the previous plan, shifted by one step, and for $\gamma > 0$ also from the zero plan, and the plan with the lower J stands.

The multiplier search of Section 4 wraps the sweeps. It quadruples the trial multiplier from one up to 10^6 and then refines the bracket with the Illinois variant of regula falsi. It stops once the information of the first input lies within 10^{-3} of the floor, or after 15 iterations.

The flights of Section 5 compute the same fixed point with L-BFGS-B, because the coordinate updates converge slowly. Every sweep shrinks the change of the plan by a factor of about 0.94, so a plan takes a median of 600 sweeps, and a step of info-CBFE takes a median of 2.7 s and up to 62 s. L-BFGS-B minimises (S29) with CasADi gradients over $|u_{k,i}| \le 1 - 10^{-6}$, with the tolerances 10^{-12} on J and 10^{-8} on the projected gradient and at most 300 iterations, from the same starts and within the same multiplier search. On the beliefs of the flights on the seeds 1000–1009 at $\sigma_f = 2$ m/s^2, the two solvers activate the constraint on the same 38% of the 200 steps, their multipliers differ by less than 3%, and their first inputs agree to 10^{-4} on 77% of the steps. On the 22 of 124 steps with an inactive constraint where they differ by more than 10^{-4}, message passing reaches the lower J on every step, by up to 0.16%, so these differences arise where L-BFGS-B stops short of the minimum. In closed loop on the seeds 1000–1099 at $\sigma_f = 2$ m/s^2, message passing attains $G = 3838$ for info-CBFE and 5124 for no-info-CBFE, against 3897 and 5129 with L-BFGS-B, with paired differences of -59 [-125, -4] and -5 [-185, 148]. The script `run_message.py` of the experiment directory runs both comparisons.

S2.4. Baselines

All controllers share the filter and the horizon $K = 6$, and the information-seeking controllers differ in how they weigh the information. No-info-CBFE minimises (S29) with $\gamma = 0$, so it keeps the

caution of the trace term but never probes. The dual MPC replaces the constraint by the fixed weight $\gamma_k = \rho/2$ on the information $\mathbb{I}_k(u_k) = \frac{1}{2}\ln\left|\Sigma_k(0.3^2 I)^{-1}\right|$ of every planned output, and L-BFGS-B minimises its objective from the same two starts as for info-CBFE. The oracle is no-info-CBFE whose parameter belief is set at every step to the true values $c = k\,\omega_{t+1}$, k, $(a, b) = (\cos\chi, \sin\chi)$ and $\eta = 1$, with zero covariance. It thus knows the fault and the wind of the coming step, but it keeps the commanded rotor plane of the model.

MPC-X minimises the nominal cost, which is (S29) without the trace term and with $\gamma = 0$, subject to the experiment-design requirement of Larsson et al. (2013). This requirement asks that the 0.95 confidence ellipsoid of θ lie within the parameters whose application cost is at most $1/\kappa$, that is, $P^{-1} \succeq \frac{\kappa}{2}\,\chi^2_{0.95}(7)\,\nabla^2 V_{\text{app}}$. We take as the application cost the goal cost of the deviation of the next position, $V_{\text{app}}(\theta) = \frac{1}{2}\|p_{t+1}(\theta) - p_{t+1}(\hat{\theta}_t)\|^2_{S_*^{-1}}$, under the first input of the nominal plan, whose Gauss–Newton Hessian is $D_1^\top S_*^{-1} D_1$ with $D_1 = \partial p_{t+1}/\partial\theta$. For P, MPC-X runs the extended Kalman filter along the plan, with the cross-covariances and the updates on both outputs, and takes the parameter block $P_{t+K}(u_\tau)$ of the joint covariance at the end of the horizon. Since the trace of a positive semidefinite matrix bounds its largest eigenvalue, the requirement

$$\operatorname{tr}\big(D_1 P_{t+K}(u_\tau) D_1^\top\big) \le \frac{2\,(0.03\ \text{m})^2}{\kappa\,\chi^2_{0.95}(7)} \tag{S30}$$

implies the matrix inequality, demands at most three times as much, and gives SLSQP a smooth constraint.

SLSQP starts from the previous plan and from the nominal plan, with exact gradients, the tolerance 10^{-10} and at most 200 iterations. When neither start meets (S30), MPC-X applies the nominal plan. The inputs change the spread in (S30) little over the horizon, because the wind drag drifts, so the requirement is either met by the nominal plan or out of reach at most steps. Relaxing the requirement to the most precise plan instead drives the quadcopter away from the target for $\kappa \ge 0.005$.

The held-out seeds 3000–3049 at $\sigma_f = 2$ m/s^2 tune the probing weight over $\rho \in \{10^2, 10^3, 3\cdot 10^3, 10^4, 3\cdot 10^4, 10^5\}$ and the accuracy over $\kappa \in \{0.003, 0.01, 0.03, 0.1, 0.3, 1\}$ by the mean of G. They select $\rho = 3\cdot 10^3$ and $\kappa = 0.003$. The accuracies 0.003, 0.1, 0.3 and 1 tie, because the nominal plan meets the requirement for $\kappa = 0.003$ and no plan meets it for $\kappa \ge 0.1$, so all four apply the nominal plan, and the smallest is taken. With $\kappa = 0.003$, the requirement binds on 0.05% of the test steps, so MPC-X acts almost as MPC on the point estimates.

S2.5. Protocol

Every controller flies the test seeds 4000–4099 at every IMU noise. Every seed draws the target, the drag coefficient, the wind, the fault and standard normal noise sequences, and the noise levels scale these sequences. All controllers and noise levels thus share the same realisations, so we report paired bootstrap 95% intervals with 10^4 resamples of the flights. The goal term of (2) scores the true position, $\frac{1}{2}\|p_k - y_*\|^2_{S_*^{-1}}$, because the position fix is ten times noisier than the tolerance. The prediction term takes μ_k and Σ_k from the one-step prediction of both outputs by the extended Kalman filter, whose Gaussian form makes this prediction exact under the filter's belief, and it contributes less than 0.1% of G for every controller.

A logged search over 72 configurations of the faults, the priors and the sensor levels on the seeds 2000–2029 proposed the setting, and the file `results/search_log.jsonl` of the experiment directory keeps every trial. We validated the candidates on the seeds 3000–3099 and also on the

seeds 1000–1099. A candidate with faults up to $\pm 115^\circ$ and a prior that trusts the compass passed on the seeds 3000–3099 but not on the seeds 1000–1099. The search then turned to priors that match the fault distribution, and the selected setting passed on both seed sets. The seeds 1000–1099 thus informed the choice of the setting, so Section 5 reports the seeds 4000–4099, which we flew once, after the setting and the tuned values of ρ and κ were fixed.

We drop every seed on which the oracle ends farther from the target than it started, at any IMU noise, because no information about the wind and the fault then makes the target reachable. This rule drops seed 4076 only, on which the oracle ends 13 m from the target, whereas no other oracle flight ends more than 4.8 m from it. With all 100 seeds, info-CBFE attains $G = 4375$ at $\sigma_f = 2$ m/s^2, against 4374 for the dual MPC, 4364 for MPC-X and 4763 for no-info-CBFE, with paired gaps of -1 [-523, 493], -11 [-1475, 890] and 388 [-468, 1063]. Seed 4076 alone accounts for this change, because it costs info-CBFE $G = 117{,}707$, against 54,555 for MPC-X and 59,289 for the oracle. On the seeds 1000–1099, the same rule drops seed 1023. The script `summary.py` of the experiment directory computes every number of Section 5 and this section.

The wall-clock time of a step covers planning and filtering. At the nominal noise, the flights ran as four parallel processes on the four Zen 5 cores of an AMD Ryzen AI 9 HX 370, so that their times compare between controllers. The other noise levels ran on all twelve cores, and Fig. 2 shows no times for them. With L-BFGS-B at $\sigma_f = 2$ m/s^2, the mean time per step of a flight has a median over the flights of 255 ms for info-CBFE, 28 ms for no-info-CBFE, 80 ms for the dual MPC and 46 ms for MPC-X. The steps of info-CBFE with an active constraint take a median of 0.62 s and at most 2.53 s, and 72% of them exceed the control period of 0.5 s, whereas no step of the other controllers takes more than 0.29 s. The code runs on Python 3.14 with NumPy 2.4, SciPy 1.18 and CasADi 3.8.

S3. Additional experiments

S3.1. State-Space Model: Integrator with Unknown Gain

The benchmark of Åström and Helmersson (1986) consists of $x_k = x_{k-1} + b\,u_k + w_k$ and $y_k = x_k$ that has to be driven to zero over $T = 50$ steps from $x_0 = 0$. The noise is zero-mean Gaussian distributed, $w_k \sim \mathcal{N}(0, \sigma^2)$, and the gain is unknown, $\theta = b \sim \mathcal{N}(b_0, 1)$. The input determines how much the next state reveals about the gain, and under $b_0 = 0$ the unknown sign of the gain turns a cautious controller off. The parameter belief is the Kalman filter $\mathcal{N}(\hat{b}_t, P_t)$, so the gain adds the variance $P_t u^2$ to the prediction, and the information of the transition node is $\frac{1}{2}\ln(1 + \ell_t(u))$ with $\ell_t(u) = P_t u^2/\sigma^2$. With the goal prior $\mathcal{N}(0, \sigma^2)$, the floor is

$$\beta_t = \tfrac{1}{2} \ln \frac{\sigma^2 + P_t\, \mathbb{E}_{\tilde{p}_t}[u^2]}{2\sigma^2} \,. \tag{S31}$$

We set $\sigma = 0.1$ and run 200 episodes for each prior mean $b_0 \in \{0, 0.5, 1, 2, 4\}$, with b drawn from its prior. The controllers and the tuning protocol are those of Section 5, but we report the root-mean-square error (RMSE) of the output, which orders the episodes as their goal cost does. Tuning on 50 held-out episodes per prior mean selects $\rho = 5$ from $\{0, 0.5, 1, 2, 5, 10, 20, 50\}$ and the accuracy 0.1 of MPC-X from $\{0.01, 0.03, 0.1, 0.3, 1\}$. The optimal dual controller follows from dynamic programming on the Bellman equation of Åström and Helmersson on a 121×121 grid without an input bound, and all other controllers keep $|u_k| \le 1$.

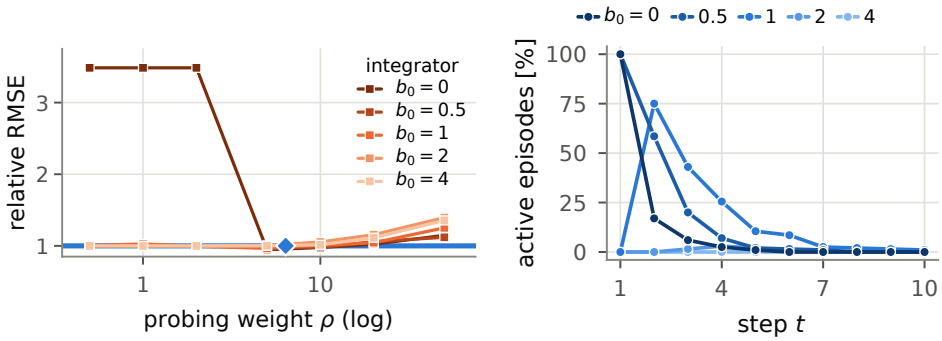


Figure S1: (Left) RMSE of the dual MPC relative to that of CBFE against its probing weight ρ, per prior mean b_0 over 200 episodes. Only $\rho = 5$ and $\rho = 10$ keep the dual MPC within 6% of CBFE under every prior mean, because smaller weights leave it idle under $b_0 = 0$ and larger weights over-probe. The blue line marks CBFE, which needs no tuning, and the diamond marks twice the median of its active multipliers. (Right) Share of the 200 integrator episodes per prior mean b_0 in which the information constraint of CBFE is active, per step. The constraint acts only in the first steps, and never for $b_0 = 4$.

CBFE comes within 6% of the RMSE of the dual MPC with the best probing weight for every prior mean, without tuning. Over all prior means, CBFE attains an RMSE of 0.116, against 0.114 for the dual MPC with the tuned $\rho = 5$ and 0.116 for MPC-X. The paired difference to the dual MPC is 0.002 [0.001, 0.004], i.e., 2.1%, and the difference to MPC-X is 0.001 $[-0.001, 0.003]$. The optimal dual controller attains 0.104, but without an input bound and by minimising the expected sum of squared outputs. Outside $5 \leq \rho \leq 10$, the RMSE of the dual MPC exceeds that of CBFE by at least 7% (Fig. S1 left). For $\rho \leq 2$, it applies no input under $b_0 = 0$, where its RMSE is 0.436, and $\rho = 50$ over-probes and exceeds the RMSE of CBFE over all prior means by 24%.

Fig. S1 (right) shows that the multiplier probes only while the gain is too uncertain for the goal. Its constraint is active on 2.0% of the steps, at the first step of every episode for $b_0 \leq 0.5$, at the second step of 75% of the episodes for $b_0 = 1$, and never for $b_0 = 4$. The active multipliers have a median of 3.2, and twice this median lies inside the window of good probing weights. Without the constraint, the cautious controller applies no input under $b_0 = 0$, because every input only adds predictive spread while the sign of the gain is unknown.

S3.2. Unknown states: light–dark positioning

A light–dark task tests whether the constraint gathers information about a hidden state whose information depends on where the agent measures (Platt et al., 2010). The agent moves along a line and measures its position with a noise that falls from the dark into the light,

$$x_k = x_{k-1} + u_k + w_k\,, \qquad y_k = x_k + v_k\,, \qquad r(x) = r_L\left(\frac{r_D}{r_L}\right)^{1-e^{-z^4}}, \tag{S32}$$

with $v_k \sim \mathcal{N}(0, r(x_k))$, $z = \min\{(x - x_L)/\ell, 0\}$, $w_k \sim \mathcal{N}(0, 10^{-4})$, the light from $x_L = 1.5$ on with $\ell = 0.8$, and the noise variance falling from $r_D = 100$ for $x \leq 0$ to $r_L = 10^{-3}$ for $x \geq x_L$.

The agent regulates x_k to $x_* = 0$ within $S_* = 0.01$ from $x_0 \sim \mathcal{N}(0, 0.25)$ over $T = 200$ steps, so the target lies in the dark.

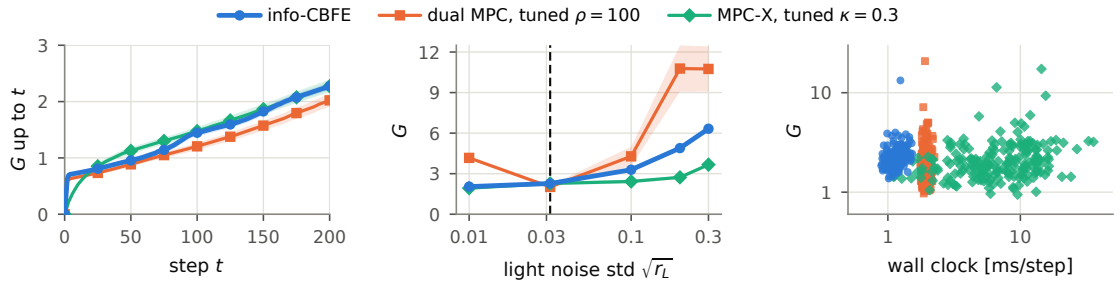


Figure S2: CBFE localises the hidden state without a tuned weight and degrades gradually when the light changes. (a) Performance accumulated over time for $\sqrt{r_L} = 0.03$. (b) Performance against the measurement noise (vertical line indicates $\sqrt{r_L}$ for (a) and (c)). (c) Performance against per-step average wall-clock time at $\sqrt{r_L} = 0.03$.

The constraint gathers information about the hidden state x_k, because the model holds no unknown parameter. A grid filter computes the exact posterior of x_t, whose mean m_t and variance V_t form the belief node, and the next output carries the state information $\mathbb{I}^x(u) = \frac{1}{2}\ln(1 + A_t/r(m_t + u))$ with $A_t = V_t + 10^{-4}$. The goal acts on the state, so the floor is $\beta_t = \frac{1}{2}\ln(A_t/S_*)$, and CBFE seeks the light whenever its belief spreads beyond the goal tolerance. The predicted variance does not depend on the plan, so a planner without an information term never visits the light. We compare CBFE with the dual MPC at $\rho = 100$, which the held-out seeds select because every weight up to $\rho = 30$ never visits the light and $\rho = 300$ spends 9% of the steps there. MPC-X requires the state precision predicted at the end of the horizon to reach $\kappa\,\chi^2_{0.95}(1)/(2S_*)$ with the held-out $\kappa = 0.3$, and we drop the ramp of Larsson et al. (2013), because the state drifts.

CBFE matches MPC-X at the nominal noise, with the lightest tail and the shortest planning time, whereas the dual MPC wins only at the noise its weight was tuned for (Fig. S2(a)). Over 200 episodes, CBFE attains $G = 2.27$, against 2.27 for MPC-X and 2.02 for the dual MPC, which thus scores 0.25 [0.12, 0.37] lower than CBFE. We score the true state in the goal term of (2), and its prediction term adds 0.49 to 0.50 for every controller, because the grid filter predicts the outputs exactly. The worst 10% of episodes reach $G = 3.81$ for CBFE, against 4.88 for the dual MPC and 5.27 for MPC-X. CBFE plans a step in a median of 1.1 ms, against 1.9 ms for the dual MPC and 6.8 ms for MPC-X, because its constraint is active on only 2.7% of the steps (Fig. S2(c)).

The tuned weight does not transfer to another noise in the light, whereas CBFE and MPC-X degrade gradually (Fig. S2(b)). At $\sqrt{r_L} = 0.01$, the dual MPC with $\rho = 100$ attains $G = 4.17$ against 2.04 for CBFE, and from $\sqrt{r_L} = 0.2$ upward it never visits the light and reaches 10.8. A noisier light favours MPC-X, which attains 3.67 at $\sqrt{r_L} = 0.3$ against 6.32 for CBFE. The floor of CBFE asks for the precision S_* even when one measurement in the light cannot deliver it, so CBFE spends 5.8% of the steps in the light, against 2.2% for MPC-X, whose requirement is looser at $\kappa = 0.3$.

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